A Differentiable Pencil-Beam-Prior Residual 3D U-Net with a Batched GPU Engine for Beamlet-Level Proton Dose Prediction on CT (DoseRAD2026 Proton-CT Task)

Kai Wang, Meixu Chen, and Rui Yang

Department of Radiation Oncology, University of Colorado Anschutz Medical Campus, Aurora, CO, USA
Corresponding author: Kai Wang, `kai.2.wang@cuanschutz.edu`

Abstract. We predict the Monte-Carlo (MC) dose of every proton beamlet by correcting an analytical Hong pencil-beam prior with a residual 3D U-Net. Five differentiable channels are computed per beamlet from the CT-derived stopping power: density, skin-entry water-equivalent path length (WEPL), the pencil-beam prior itself (ported to a self-contained GPU implementation, r=0.995 against the pyRadPlan reference, so the deployed system computes its own prior with no planning-system dependency), lateral distance, and energy. The prior supplies ∼95% of the dose; in ablation, correcting a systematic WEPL over-count adds +2.4 γ(1%/1 mm) and the prior a further +4.4, lifting a plain U-Net from 89.2 to 97.0 on the development fold. Under 5-fold cross-validation on the 75 public patients the model reaches plan γ1%/1 mm = 96.7 ± 1.0, the strongest of our four task systems. Deployment uses a batched engine built on a geometric identity; pencil-beam-spot rays within one beam are exactly parallel, so per-beam WEPL collapses to one orthographic beam's-eye-view cumulative sum (388× over ray-marching), with bucket-batched network forwards (exact by same-shape bucketing) and streamed compressed output: 0.185 s per dose map at unchanged gamma. On the hidden preliminary test set the full-data submission scored γ = 95.60, per-beam MAE 0.0082, IDD 0.0047, stratified MAE 0.0077, DVH 0.83, well within the 500 s gate.

Keywords: proton dose prediction · pencil beam · differentiable physics · residual learning · WEPL

1 Introduction

Proton dose calculation is dominated by range: the Bragg peak position is set by the integrated stopping power along the beam path, so small upstream density errors displace large local doses. Deep engines such as DoTA [1] reach MC accuracy at millisecond latency but treat transport as an implicit target. We factor the problem instead: an analytical, differentiable physics operator $\mathcal{C}$ turns the

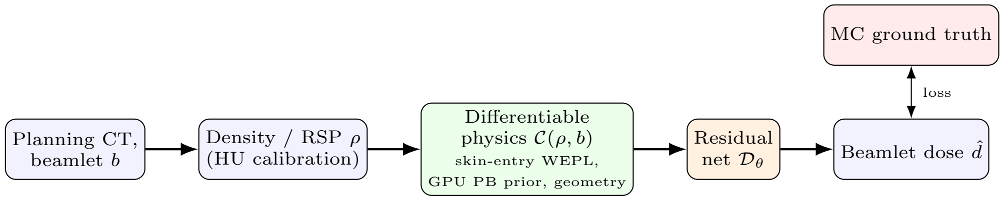

Fig. 1. Proton-CT pipeline: per beamlet, differentiable channels (skin-entry WEPL, GPU pencil-beam prior) from the CT stopping power; the U-Net learns the residual to MC.

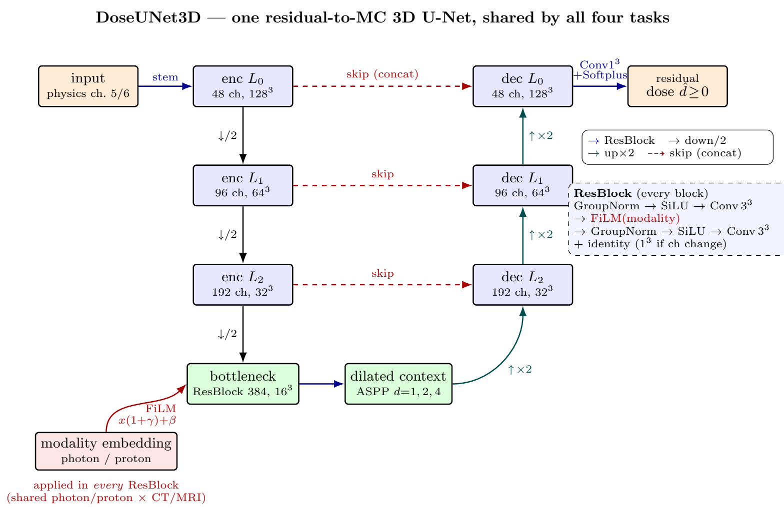

Fig. 2. The dose network $\mathcal{D}_\theta$ (DoseUNet3D): residual-to-MC 3D U-Net (channels 48/96/192/384), dilated-context (ASPP) bottleneck, Softplus head, FiLM modality conditioning in every ResBlock; the photon and proton networks share this architecture but differ in input channels.

CT-derived density and the beamlet descriptor into dose-shaped channels, including a full analytical pencil-beam prior [2] that already supplies ∼95% of the answer; and a residual 3D U-Net $\mathcal{D}_\theta$ corrects the remainder (lateral penumbra, scatter tails, heterogeneity effects the pencil beam misses). Two properties of this design matter in practice. First, data efficiency: with most of the dose supplied analytically, the network learns a small, smooth correction, and 75 patients suffice for $\gamma 1\%/1\,\mathrm{mm} \approx 97$. Second, deployability: we port the pencil-beam engine to a self-contained GPU implementation, so the deployed container computes its own prior: no external treatment-planning system at inference; and we exploit beam geometry to batch the entire physics pipeline (Sec. 3.4). This report details the Proton-CT system; the same framework serves our submissions to all four DoseRAD2026 tasks.

2 Methods

2.1 Data

We use only the official DoseRAD2026 training set [5] (75 patients: 39 thorax, 36 abdomen; SynthRAD2025 cohort [6]): per patient a planning CT, beam parameters, and per-beamlet Monte-Carlo dose-to-medium in absolute Gy (Geant4 full particle transport; the treatment plans are generated with matRad [3]) on the native $1{\times}1{\times}3$ mm grid; several hundred to $\sim$a thousand beamlets (energy $\times$ spot) per patient. No external data are used for this task. Development used 5-fold cross-validation over all 75 patients; **the final submitted model is retrained on all 75**, with the challenge leaderboard as the held-out validation signal.

2.2 Model

$\hat{d} = \mathcal{D}_\theta(\mathcal{C}(\rho, b))$ with ρ treated as relative stopping power from the official HU calibration. $\mathcal{C}$ assembles **five differentiable channels** per beamlet: (1) *density/RSP* ρ; (2) *water-equivalent path length* (WEPL): a per-voxel integral of density from just upstream of the skin to the voxel. WEPL is the single most range-determining input, and we compute it with a corrected per-voxel scheme rather than a naive beam's-eye-view fan, which we found to carry a systematic over-count worth $-2.4\,\gamma$ (Table 2); (3) the *Hong pencil-beam prior* [2]: the full analytical beamlet dose. We ported the reference implementation (pyRadPlan) to a self-contained GPU version that reproduces it to $r = 0.995$ at $\sim$40 ms/beamlet, and finetuned the residual network on this GPU prior so that cached-prior training is identical to on-the-fly inference (train = deploy); (4) *lateral distance* from the beamlet axis (penumbra, multiple-Coulomb scatter); (5) *energy* (nominal Bragg-peak depth). The per-ray *skin-entry* rule applies to the WEPL and prior: depth accumulates only from the first voxel with $\rho > 0.05\,\mathrm{g/cm^3}$; the prior is masked strictly upstream: no dose before the body, internal cavities untouched. Figure 3 shows the deployed model's complete residual-learning chain; note how small and structured the residual is once the pencil-beam prior is subtracted.

2.3 Training

Loss: high-dose-weighted L_1 to MC (high-dose fraction 0.1, weight 10) plus bounded penumbra/interface/lung terms; AdamW (lr 10^{-4} scratch, 2×10^{-5} finetune), EMA, gradient clipping; patches 32×128^2 (thin along the beam, wide laterally), batch 2–4. The network trains $\sim$120k steps on cached channels, then a finetune on the GPU pencil-beam prior (train = deploy) and a final convention-consistency finetune matching the deployed batched WEPL engine (Sec. 3.4; the orthographic per-beam convention differs from the per-voxel march by a mean $|\Delta\mathrm{WEPL}|$ of $0.18\,\mathrm{g/cm^2}$, and a short finetune makes the network robust to both conventions). Reproducibility: fixed seeds, YAML configs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

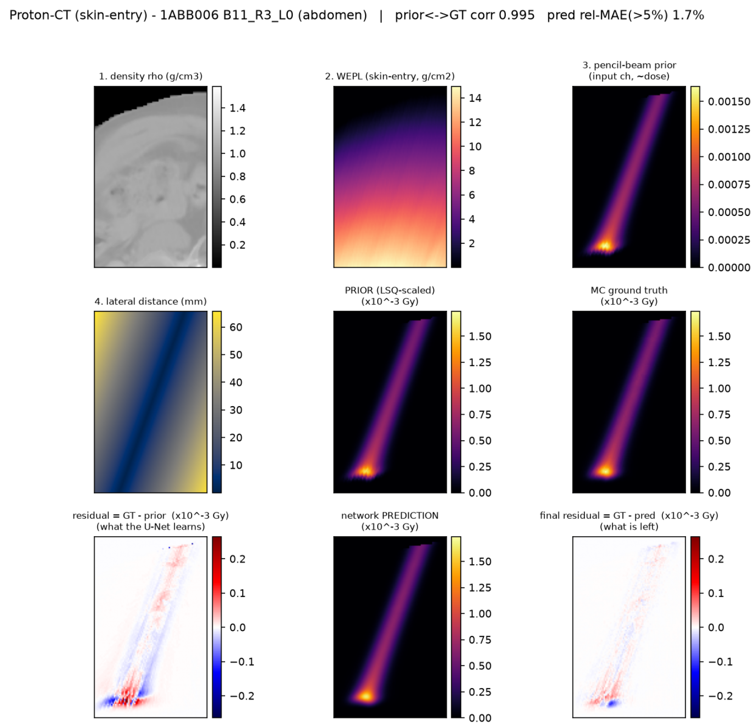

Fig. 3. Residual-learning chain of the deployed skin-entry model (abdomen, peak-dose beamlet): inputs (density/RSP, skin-entry WEPL, lateral distance) → GPU pencil-beam prior (already in Gy, ∼95% of the dose) → MC ground truth → residual (GT−prior) → prediction → final residual (GT−pred); both residual panels share one symmetric window.

2.4 Evaluation

Internal: plan-level local γ [4] (pymedphys, 1%/1 mm, ≥10% Rx cutoff, byte-identical to the vendored official code), per-beamlet masked MAE, official IDD and stratified-MAE reimplementations, and a frozen 16-patient held-out cohort gating every deployment change (*no accuracy loss, no speed loss*). Runtime: official score $T = t_{\mathrm{fix}} + N \cdot t_{\mathrm{dose\,map}}$, gate 500 s/job, double-weighted; all engine work was profiled at job scale under a strict 14.5 GB platform memory budget (our development GPU has 32 GB, which silently masks platform OOM; job-scale replicas at the platform budget caught two such failures before submission).

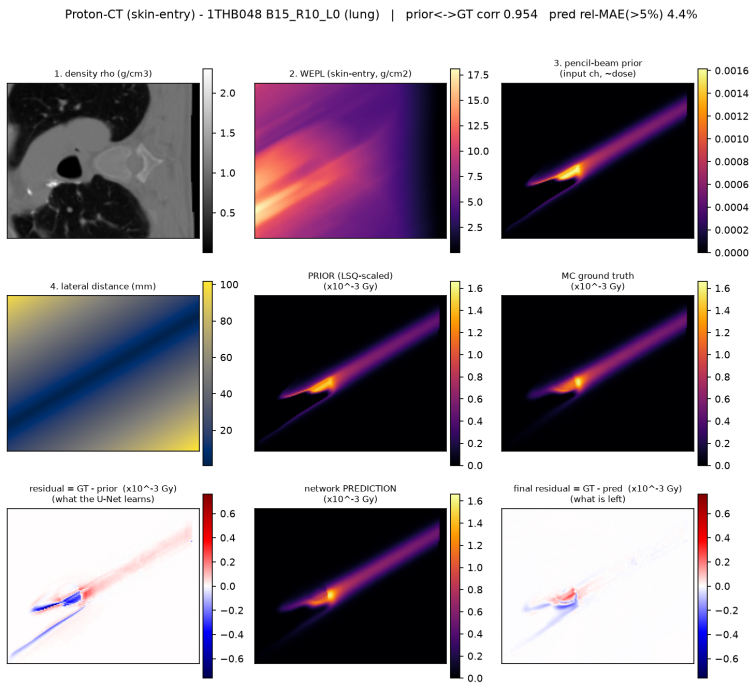

Fig. 4. As Fig. 3, lung case: the pencil-beam prior degrades most in low-density lung (lateral heterogeneity), which is where the residual network contributes.

3 Results

3.1 Accuracy: cross-validated and hidden-test

Per-fold CV $\gamma 1/1$: 96.7/97.7/96.9/94.9/97.5 (mean $\mathbf{96.7} \pm \mathbf{1.0}$); the strongest of our four task systems, reflecting the strength of the analytical prior. Development fold ($n = 16$): $\gamma 1/1 = 97.0 \pm 2.3$ (abdomen 98.2, lung 95.8), $\gamma 3/3 = 99.9$; per-beamlet masked MAE $0.89 \pm 0.22\%$. Table 1 adds the official hidden-test metrics of the full-data submission.

3.2 Ablations: the prior does the heavy lifting

Table 2 isolates the two Proton-CT-specific levers on the development fold. A plain U-Net on uncorrected channels reaches 89.2; fixing the systematic WEPL over-count adds +3.4; adding the pencil-beam prior a further +3.1; finetuning the residual on the deployable GPU prior (train=deploy) closes the chain at 97.0. The analytical prior is not a warm start but the backbone: it supplies $\sim$95% of the dose, and the network's role is confined to the residual the analytical model cannot express.

Table 1. Proton-CT results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission on the preliminary test set (August 2026). Gate: 500 s/job.

Metric	Value
CV plan γ1%/1 mm (5-fold)	96.7 ± 1.0
Hidden-test per-beam masked MAE	0.0082 ± 0.0025
Hidden-test IDD curve distance	0.0047 ± 0.0025
Hidden-test stratified plan MAE	0.0077 ± 0.0028
Hidden-test plan γ1%/1 mm	95.60 ± 3.33
Hidden-test DVH clinical score	0.83 ± 0.35
Hidden-test runtime per job	191.7 s (batched engine: 0.185 s/map, 57 s/309-map job)

Table 2. Proton-CT ablation (plan γ1%/1 mm, development fold, $n = 16$).

configuration	ALL	abd	lung
no prior (WEPL uncorrected)	89.2	90.7	87.6
no prior + WEPL-fix	92.6	93.1	92.2
+ pencil-beam prior	95.7	96.9	94.5
+ GPU-PB finetune (train=deploy, deployed)	**97.0**	98.2	95.8

3.3 Saturation: why extra training signals are flat here

On the photon task, multi-modal joint training (one network over real- and synthetic-CT inputs) gains $+2.8\,\gamma$ cross-validated. The *identical* experiment on Proton-CT is flat (96.7 ± 0.9, mean per-fold change -0.01), and the real-CT-ceiling probe agrees (-0.6). This is the expected behaviour of a residual architecture whose prior already supplies most of the signal: the residual is small (penumbra and scatter tails) and the network operates near the floor set by the prior's fidelity and label noise. The contrast with photons is a useful design rule: *invest extra training signal where the analytical prior is weak.*

3.4 Error analysis: the lung deficit is a systematic over-prediction

Aggregate gamma hides error structure, so we analyzed the failing voxels directly. In lung, the failures are not symmetric noise: 73.7% of gamma-failing voxels are *over*-predictions (binomial $p \approx 3 \times 10^{-13}$ against symmetry); the model systematically deposits slightly too much dose in low-density lung, plausibly inherited from the pencil-beam prior's handling of lateral heterogeneity, which the residual network under-corrects. A post-hoc lung-region deflation corrected the bias but degraded net lung gamma (the correction helps failing voxels and harms passing ones), so the fix belongs in training, not post-processing; we report the diagnosis as a target for the next iteration. Abdomen shows no such structure.

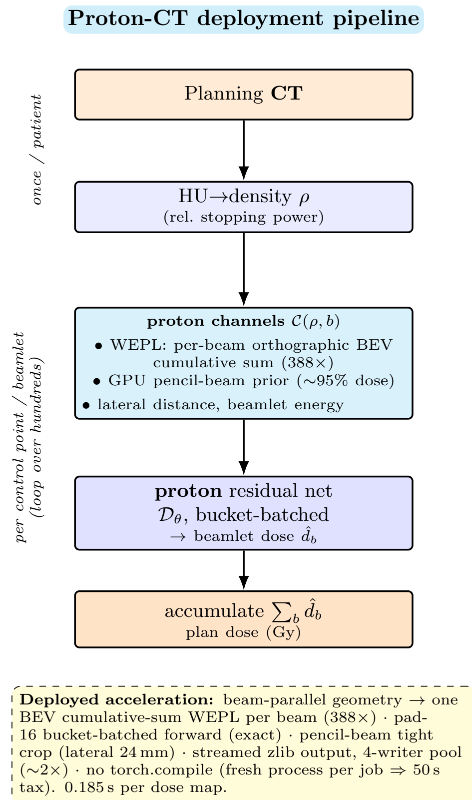

Fig. 5. The deployed Proton-CT container: per patient, the density stage runs once; the differentiable-channels → residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

3.5 Runtime engineering: a batched proton engine

Proton jobs write one dose map per beamlet on the native $1{\times}1{\times}3$ mm grid (tens of GB per job) so both compute and write paths matter. Our engine (Table 3) rests on a geometric identity: within one beam, pencil-beam-spot rays are *exactly* parallel (the virtual source translates with the spot), so the per-beam WEPL of all beamlets collapses to a single orthographic beam's-eye-view cumulative sum; 388× faster than per-voxel ray-marching, and matched to training by the convention finetune of Sec. 2.3. Downstream, beamlet crops are padded into shape buckets and batch-forwarded (plain GroupNorm is per-sample, so same-shape bucketing keeps batching *exact*; we abandoned a masked-GN variant after it cost $13\,\gamma$), and finished frames stream through a 4-worker zlib-compression pool that overlaps writing with GPU compute ($\sim$2× on the otherwise write-bound path). End-to-end: 0.185 s/map on the evaluation platform (57 s for a 309-

Table 3. Proton-CT engine levers (all accuracy-gated; measured on the evaluation platform or job-scale replicas at the 14.5 GB platform memory budget).

Lever	Effect	Accuracy impact
Orthographic BEV cumulative-sum WEPL	388× WEPL build	convention finetune
Pad-16 bucketed batched forward	part of 1.7–2× e2e	exact (same-shape GN)
Pencil-beam tight crop (lateral 24 mm)	91.5→77 ms/beamlet	γ unchanged
Streamed zlib output, 4-writer pool	∼2× (write-bound path)	lossless
Sub-cutoff quantisation of output	smaller writes	γ preserved (gated)
Batched engine, end-to-end	0.307→0.185 s/map	held-out 97.51 vs. 97.50
Rejected: `torch.compile`	+50 s/job tax, recompile storms	—

map job), 1.7–2.0× over our previous per-beamlet engine at unchanged accuracy (held-out γ 97.51 vs. 97.50). Two platform findings for future participants: (i) the evaluation service starts a fresh process per job, so `torch.compile` pays its full ∼50 s compilation tax every job; at 250–310 maps/job the tax exceeds the ∼1.7× steady-state kernel gain, and multi-grid jobs recompile repeatedly (one measured job: 238 s compiled vs. 110 s eager); our engine therefore ships *without* compile; (ii) per-beamlet crop tightening (lateral 24 mm) is a free 1.2× (91.5→77 ms/beamlet) with gamma unchanged.

Memory engineering. Two of our engine submissions failed on the platform with out-of-memory errors that our 32 GB development GPU had silently masked; the fixes; per-beam chunked streaming (memory $O(\text{chunk})$ instead of $O(\text{plan})$), an explicit batched-forward voxel cap tuned to the platform, and allocator `expandable_segments`; were validated by replaying the largest test-scale patients under a hard 12.8 GB cap before resubmission. We recommend this cap-replay protocol to any team whose development hardware out-classes the evaluation instance.

4 Discussion

Proton-CT is the clearest demonstration of the physics-prior residual design: the analytical pencil beam supplies ∼95% of the dose, the U-Net corrects only what the analytical model cannot express, and the result (96.7 ± 1.0 cross-validated) is our strongest task system with the least data hunger. The deployable GPU port of the prior removes the planning-system dependency that keeps many published engines from running standalone, and the batched engine shows that careful use of beam geometry buys ∼2× wall-clock at exactly zero accuracy cost. **Limitations and generalizability.** Single benchmark (abdomen/thorax), one machine model; the analytical prior assumes the machine's beam model, so transferring to another machine requires re-fitting the prior (the network, learning only a residual, should transfer more easily). Ablation deltas are single-fold ($n = 16$); headline accuracy is the 5-fold CV. Lung remains the harder site (95.8 vs. 98.2): the pencil beam handles lateral heterogeneity worst exactly there.

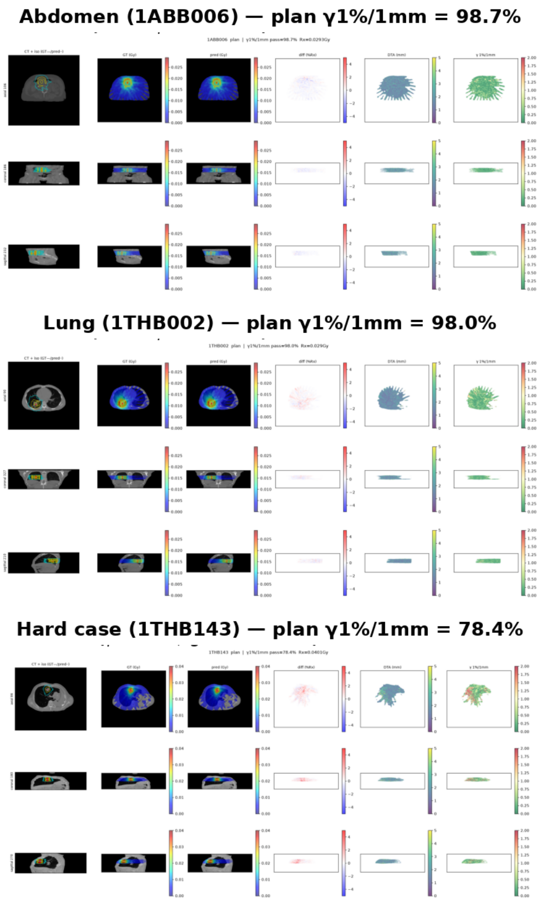

Fig. 6. Qualitative Proton-CT results on three cases spanning the difficulty range: abdomen (1ABB006, plan $\gamma 1/1 = 98.7$), lung (1THB002, 98.0), and the geometrically hard case 1THB143 (78.4; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local γ1%/1 mm.

Future work. (i) *Fix the lung over-prediction at the source*: our error analysis (Sec. 3.4) localizes a systematic bias that post-processing cannot remove; a lung-aware loss asymmetry or a heterogeneity-corrected prior variant (e.g. subdividing the pencil beam laterally in low-density regions) is the natural next training-side intervention. (ii) *Machine transfer*: the residual design should transfer across machines by re-fitting only the analytical prior; validating this on a second beam model would substantiate the framework's main practical claim. (iii) *DVH-aware selection*: as on the photon task, our DVH clinical score lags our gamma rank; internalizing the official DVH pipeline for checkpoint selection is cheap and likely valuable. (iv) *Gradient-based planning*: the engine is differentiable down to beamlet weights and energies, making it a candidate physics layer for direct proton spot-map optimization [7,8]; with the batched engine at 0.185 s/map, an optimization loop over a full spot map is already interactive-rate.

5 Author Contributions (CRediT)

Kai Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu Chen:** Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing.

6 Other Information

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